Browser Doctor AI

Product Launch Documentation & Publishing Guide

PREPARED FOR

Chrome Web Store Deployment

VERSION

1.0.0 (Manifest V3 MVP)

DATE

July 2026

STATUS

Production-Ready

Executive Summary

Browser Doctor AI is a modern, lightweight, and event-driven Google Chrome extension built to optimize browser resources without degrading active user experiences. Unlike traditional browser managers that consume high background memory or poll system metrics constantly, Browser Doctor AI uses an intelligent estimation logic to measure page complexity (DOM size, mutations, media objects) and reactively suspends inactive background tabs using Chrome's native discard mechanism.

Designed with a minimalist, premium aesthetic inspired by applications like Arc Browser, Raycast, and Linear, this extension blends into the web browser as an integrated OS-level utility.

Core Features

Floating Health Bubble

A non-intrusive, draggable pill widget injected into the page via a Shadow DOM. Displays a real-time health score from 0–100 and status color dot. Semi-transparent when idle, expanding slightly upon hover.

Memory Scoring Engine

Calculates per-tab weights using DOM node count, media counts (images/videos), active WebGL/canvas contexts, iframes, script declarations, and live DOM mutation frequency.

Mini Diagnostics Panel

Opens directly on the page when clicking the floating bubble. Offers a quick, lightweight snapshot of heavy tabs, duplicates, idle tabs, and overall rating.

Full Performance Dashboard

A dedicated, full-page dashboard compiled with React, TypeScript, and Tailwind CSS. Contains detailed metrics, a SVG performance timeline graph, diagnostic lists, and settings options.

How to Use & Manage

Operating Browser Doctor AI is designed to be frictionless, requiring no initial setup or account registration. The extension begins running diagnostics immediately after installation.

1. Finding the Floating Bubble

Once loaded, navigate to any webpage. The green, blue, yellow, or red pill is anchored to the top-right corner by default. You can drag and position it to any quadrant. The position is stored across page loads.

2. Triggering One-Click Optimizations

When the browser health drops (e.g. below 70%), you can open the mini-panel or full dashboard to run optimizations:

- **Close Duplicates:** Instantly compares all open tabs' URLs, matching titles and domains, keeping the first occurrence and closing redundant tabs.
- **Suspend Idle Tabs:** Recovers memory from tabs inactive for >15 minutes. Under the hood, this triggers the native `chrome.tabs.discard()` API, which fully unloads the tab's memory while keeping its tab header in Chrome for instant reload when clicked.

3. Customizing settings

Go to the Settings panel inside the dashboard to toggle preference configurations:

> **Key Customizations Available**
>
> Adjust bubble size (Small, Medium, Large) to fit high DPI displays; modify the notification toggle; adjust health score threshold sliders; and toggle the auto-launch dashboard feature on startup.

Competitive Advantages

Most browser managers suffer from design and performance issues. Here is how Browser Doctor AI outclasses competing products:

Metric / Quality	Standard Extensions	Browser Doctor AI
Host Page Footprint	High (bundles React/jQuery directly onto user pages)	Minimal (< 12KB vanilla script, isolated in Shadow DOM)
CPU Overhead	Heavy polling intervals querying tabs continuously	Event-driven listeners and throttled mutation observers
Memory Suspension	Replaces tabs with heavy redirect pages	Triggers native Chrome memory discard seamlessly
Design Aesthetics	Generic, cluttered, and outdated UI grids	Premium glassmorphism layouts matching modern OS tools

Web Store Publishing Checklist

Before submitting to the Chrome Web Store Developer Console, complete the following pre-publishing verification steps:

1. **Confirm Build Size:** Verify the `content.js` is less than 50 KB (our build is ~11 KB).
2. **Permissions Check:** Ensure only `storage`, `tabs`, and `notifications` are requested to satisfy the Store's "Least Privilege" security review.
3. **Prepare Asset Pack:** The store requires promotional images (440x280 small tile, 1400x560 marquee) and the generated 16x16, 32x32, 48x48, and 128x128 icons inside the `dist/` folder.
4. **Privacy Policy URL:** Chrome requires a policy link since the extension checks tab URLs to identify duplicate pages. Data is strictly processed locally on-device.